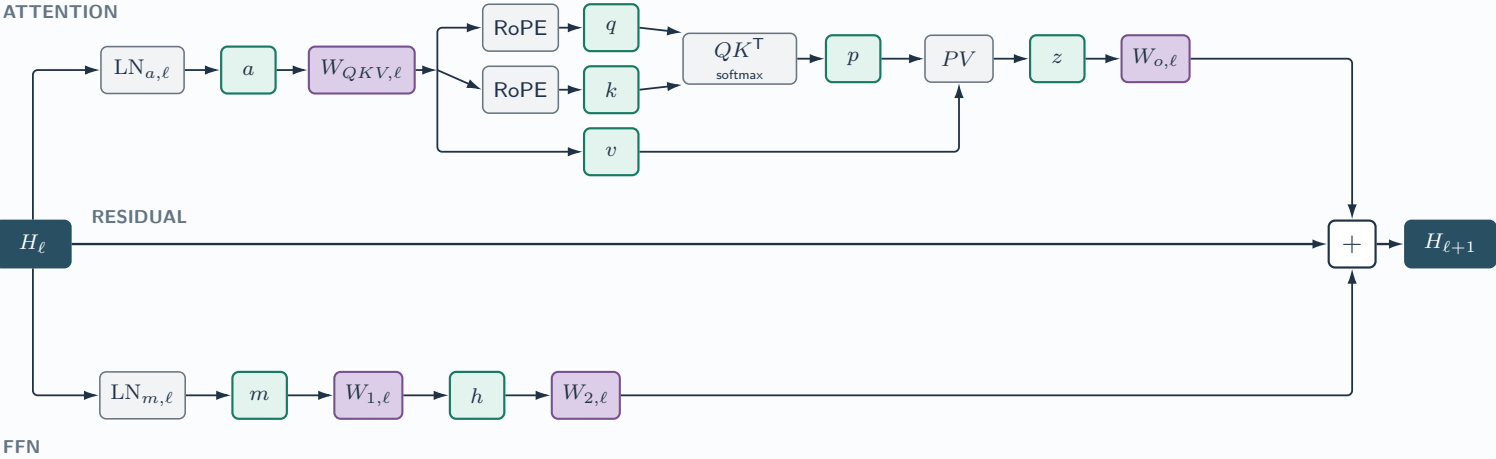


Shared Pythia block

ATTENTION



Sparsification ladder

Non-linearity   ● GeLU   ● ReLU   ●  $G_{+,\kappa}$    ●  $G_{\pm,\kappa}$    ● none

| Labels |          | Intervention                                                                        |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |          | $R_{\text{model}}^{\text{max}}$ at $T = 2,048$ |       |       |
|--------|----------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|----------|------------------------------------------------|-------|-------|
| Step   | ID       | $h$                                                                                 | $m$                                                                                 | $a$                                                                                 | $z$                                                                                 | $q$                                                                                 | $k$                                                                                 | $v$                                                                                 | Pressure | 14M                                            | 70M   | 410M  |
| 1      | A0       |    |    |    |    |    |    |    | none     | 0.0%                                           | 0.0%  | 0.0%  |
| 2      | A1-H     |    |    |    |    |    |    |    | none     | 4.3%                                           | 12.4% | 24.9% |
| 3      | A1-H-L1  |    |    |    |    |    |    |    | L1N      | 4.3%                                           | 12.4% | 24.9% |
| 4      | A1-H-OL1 |    |    |    |    |    |    |    | OL1      | 4.3%                                           | 12.4% | 24.9% |
| 5      | A4       |    |    |    |    |    |    |    | none     | 12.8%                                          | 37.1% | 74.8% |
| 6      | A4-OL1   |    |    |    |    |    |    |    | OL1      | 12.8%                                          | 37.1% | 74.8% |
| 7      | A7       |    |    |    |    |    |    |    | none     | 30.0%                                          | 49.4% | 87.2% |
| 8      | A7-OL1   |  |  |  |  |  |  |  | OL1      | 30.0%                                          | 49.4% | 87.2% |